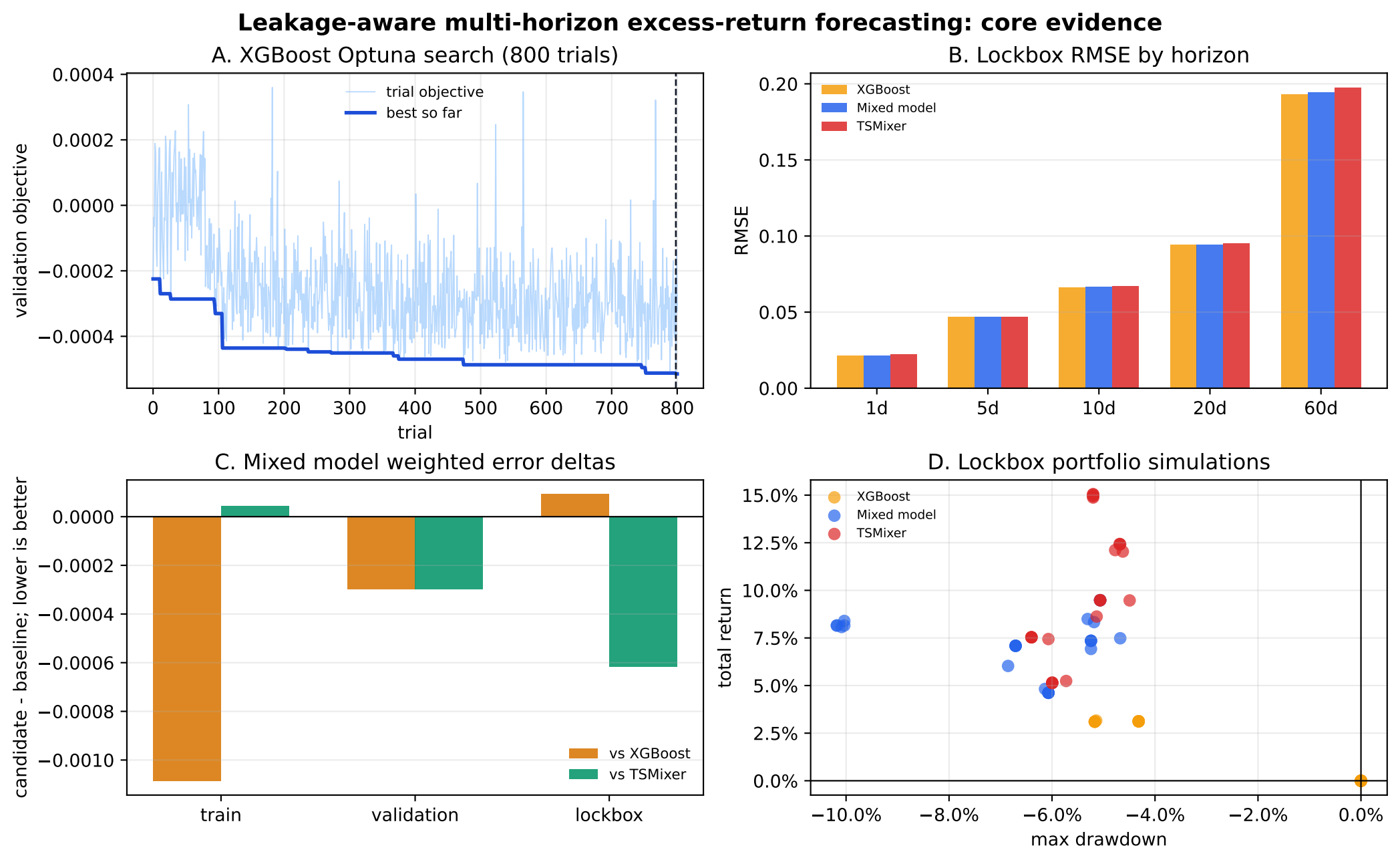
Leakage-aware multi-horizon excess-return forecasting: core evidence
A. XGBoost Optuna search (800 trials)
validation objective
trial
trial objective
best so far
B. Lockbox RMSE by horizon
RMSE
XGBoost
Mixed model
TSMixer
1d
5d
10d
20d
60d
C. Mixed model weighted error deltas
candidate - baseline; lower is better
vs XGBoost
vs TSMixer
train
validation
lockbox
D. Lockbox portfolio simulations
total return
max drawdown
XGBoost
Mixed model
TSMixer